

# Agentic AI and Multi-Agent System

Date = 6th Aug 2026

Any GenAI App/2 =>

Cutoff date  
Dec 2023 -> 2026

Input by user: What is gradient descent?

ML = Model  
Optimization  
tech.

LLM: Llama 3.1 8B

Generates a response from its prompt and training knowledge

General Ans  
Correct

Plan a 2-day trip to Jaipur next weekend from August 8 to August 9, 2026 and give me an estimated budget for the trip.

LLM: Llama 3.1 8B

Hallucination

Input  
date when  
m ->  
w ->  
w ->

request

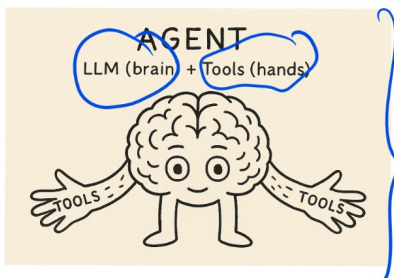
Tools -> DuckDuckGo / Tavil  
Weather API

Observation

LLM: Llama 3.1 8B

Final Answer

Understanding of logic -> plan



Agent

=> Individual software program.  
↳ perform a specific task => General a human does.  
With human interaction  
↳ Goal => Individual works

## Agent Example

Imagine a Travel Planning Agent. ✓✓

User Request --> Plan a 2-day trip to Jaipur next weekend and give me an estimated budget.

### What an Agent Does

- Instead of answering immediately, it may: ✓
- Understand the goal. ✓
- Check the dates. ✓
- Search for hotels. ✓
- Look up tourist attractions. ✓
- Estimate travel costs. ✓
- Create an itinerary. ✓ → final Ans

### Internal Flow:

User Goal



LLM reasons: "I need hotel prices and attractions"



Use Hotel Search Tool



Use Maps/Attractions Tool

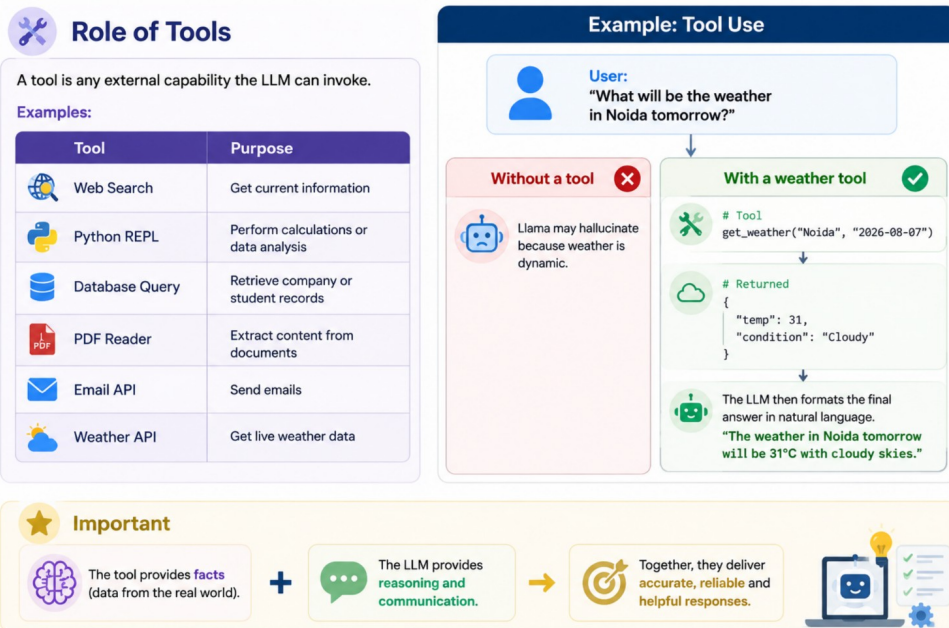


Use Budget Calculator Tool



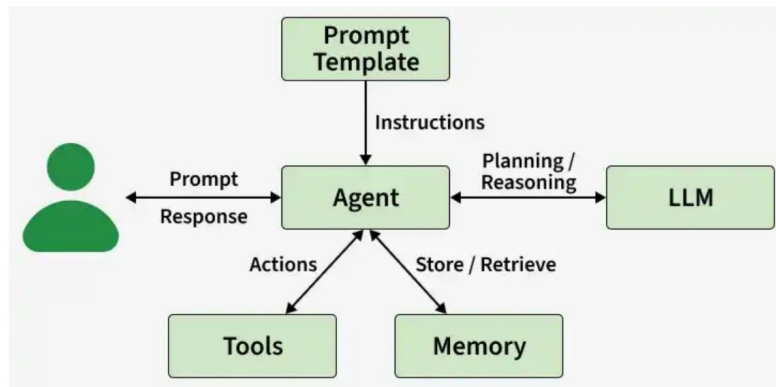
Generate final itinerary

Notice that the LLM is not only generating text; it is deciding which action to take next.



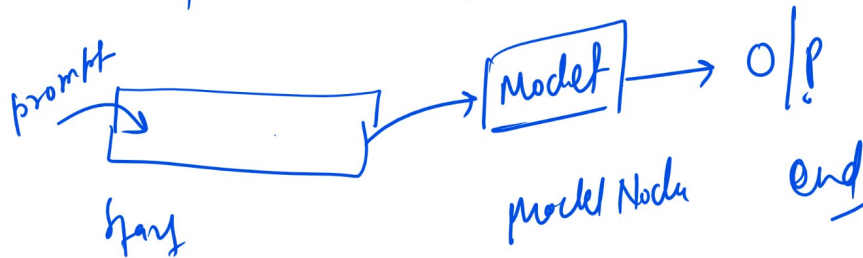
Agent → Autonomous  
 ⇒ Ask

{  
 → Perceiving  
 → Reasoning  
 → Memory  
 → Learning  
 }



Langchain

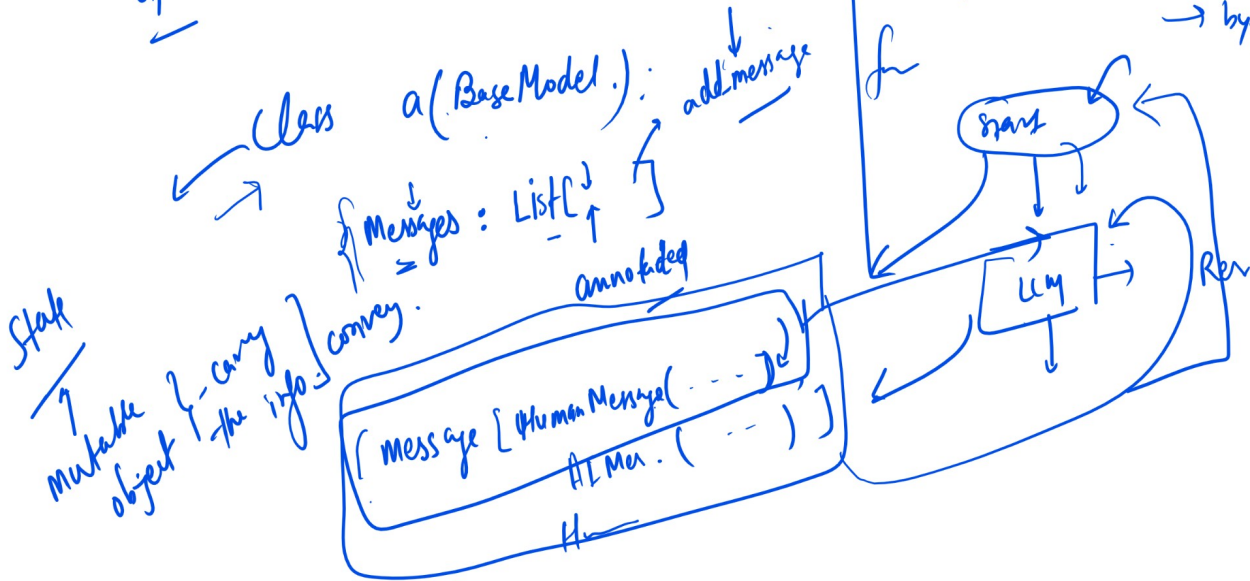
chatbot { I → model → Ans }



Hydantie  
State

chatbot

{ message. ⇒ multi-turn conversation }  
 → bye



State  
 mutable object { carry the info }